

Project FORESIGHT – Demand & Inventory Intelligence

Project Name: Project FORESIGHT

Report Type: Inventory Risk Analysis Report

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Tools Used: Python, Pandas, Scikit-learn, Streamlit

1. Introduction

This report presents the inventory risk analysis generated using the demand forecasting model. The objective is to identify products that are likely to experience stock shortages or excess inventory, enabling proactive inventory planning and improved supply chain efficiency.

The analysis is based on:

- Forecasted Demand
- Current Inventory
- Reorder Point
- Incoming Stock (On Order)

2. Risk Assessment Methodology

The inventory risk score is calculated by comparing forecasted demand with available stock.

1. Stockout Risk

Occurs when:

$$\text{Forecast Demand} > \text{Available Inventory}$$

2. Overstock Risk

Occurs when:

$$\text{Available Inventory} > \text{Forecast Demand}$$

3. Risk Levels

Risk Score	Risk Level
0 – 39	Low
40 – 69	Medium
70 – 100	High

3. Stockout Risk Analysis

Products with forecast demand greater than available stock.

SKU ID	Product	Forecast Demand	Available Stock	Risk Level
101	Office Chair	220	150	High
108	Gaming Chair	180	120	High
115	Study Table	140	100	Medium
123	Bookshelf	95	70	Medium

Business Impact

- Lost sales opportunities
- Poor customer satisfaction
- Emergency procurement costs
- Supply chain disruptions

4. Overstock Risk Analysis

Products where inventory exceeds forecast demand.

SKU ID	Product	Available Stock	Forecast Demand	Risk Level
205	Desk Lamp	300	90	High
212	Wall Shelf	240	110	High

218	Pen Holder	180	80	Medium
225	Storage Box	150	90	Medium

Business Impact

- Higher warehouse costs
- Slow inventory turnover
- Risk of product obsolescence
- Reduced cash flow

5. High Risk Products

The following products require immediate attention.

Product	Risk Type	Risk Score	Priority
Office Chair	Stockout	95	Critical
Gaming Chair	Stockout	90	Critical
Desk Lamp	Overstock	92	Critical
Wall Shelf	Overstock	88	High

Observation

- High-demand products should be reordered promptly.
- Overstocked items should be cleared through promotions or adjusted procurement.

6. Risk Distribution

Risk Level	Number of Products	Percentage
Low	157	78.5%
Medium	28	14.0%
High	15	7.5%

Interpretation

- Most products have healthy inventory levels.
- A smaller group requires immediate action to prevent business impact.

7. Business Recommendations

a. For Stockout Risk

- Increase reorder quantities for high-demand products.
- Review safety stock levels regularly.
- Place purchase orders before expected demand peaks.
- Monitor inventory daily for fast-moving products.

b. For Overstock Risk

- Launch promotional campaigns to reduce excess stock.
- Bundle slow-moving products with popular items.
- Reduce future purchase quantities for low-demand products.
- Reassess reorder points periodically.

c. Operational Recommendations

- Retrain the forecasting model using recent sales data.
- Implement automated inventory alerts.
- Monitor supplier lead times.
- Review inventory KPIs weekly.

8. Key Risk Metrics

KPI	Value
Total Products	200
Healthy Inventory	157
Medium Risk Products	28
High Risk Products	15
Average Risk Score	32.4
Forecast Accuracy (R ²)	0.94

9. Conclusion

The inventory risk analysis highlights products that require attention due to potential stock shortages or excess inventory. By combining demand forecasts with current inventory data, the organization can improve inventory planning, reduce operational costs, and maintain better product availability.

Regular monitoring of risk scores, coupled with timely business actions, will help minimize stock-related issues and support more efficient supply chain operations.